

Zain Ul Abidin

Software Engineer

Full-Stack Software Engineer with 5+ years of experience building production-grade products across Fintech, Healthcare, CRM domains. Adept at architecting end-to-end solutions from scalable backend APIs and microservices in Node.js and Laravel, to responsive, pixel-perfect frontends in Vue/Nuxt and React/Next.js. Experienced in AI/LLM integration using Python designing custom LLM workflows, automation pipelines, and intelligent feature layers. Familiar with DevOps tooling including AWS Lambda, Redis, PM2, and Git-based CI/CD workflows, with practical experience in development and deployment environments. Works fluently across relational (PostgreSQL, MySQL) and non-relational (MongoDB, Redis) databases. Delivered robust platforms spanning banking dashboards, merchant portals, accounting systems, payment gateway integrations, and healthcare data pipelines with a strong emphasis on security, scalability, and clean system design.

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🔗 <https://github.com/zaini019>

SKILLS

LaravelNode.jsVue.jsReact.jsREST APIs / GraphQLMongoDBPostgresMySQLPHPJavascriptAWS

Vuetify / Quasar / Tailwind CSSPinia / Vuex / ReduxMicroservicesCI/CD (Github Actions)Third-party Integration

PythonFastAPI / DjangoLLM IntegrationAI Workflow AutomationSTT / TTS / VAD Systems

WORK EXPERIENCE

Software Engineer

Developers Studio 06/2023 - Present

Achievements/Tasks

- Delivered end-to-end full-stack solutions across AML compliance, open banking, Binance trading, and B2B accounting platforms encompassing database architecture, API design, and frontend implementation using modern JS frameworks.
- Architected and implemented scalable RESTful and GraphQL APIs with Node.js, Laravel, and FastAPI enforcing JWT authentication, RBAC, and Sequelize ORM for optimized data management and granular access control.
- Engineered Redis-backed asynchronous task queues using Bull and Kue, handling high-volume data ingestion pipelines and improving overall system scalability and responsiveness under production load.
- Integrated real-time bidirectional communication via Socket.IO in Node.js, powering live GPT-driven client interactions and enabling low-latency, event-driven data flow across distributed services.
- Constructed LLM-powered agentic workflows and AI pipelines incorporating LangChain, Ollama, Whisper, and YOLO to deliver production-grade voice systems (STT/TTS/VAD), computer vision features, and context-aware automation.
- Developed an IoT real-time data collection backend with Python FastAPI, seamlessly integrated with a Nuxt.js back-office management system supporting dynamic routing, API orchestration, and full CRUD operations.
- Built an automated Binance Futures trading bot leveraging Python, Selenium, and MongoDB enabling stateful execution, adaptive strategy flows, and real-time market signal processing.
- Implemented intelligent web automation systems and workflow pipelines using Python and headless browser tooling, streamlining complex business processes and reducing manual operational overhead.
- Enhanced open banking data integration efficiency by 30% through modular service design, optimized data flow architecture, and strategic refactoring of legacy integration layers.
- Integrated dynamic frontend capabilities including iframe orchestration, third-party service embedding, route guards, RBAC-driven permission handling, and session management in Vue.js and React.js with scalable state management via Vuex/Pinia/redux and responsive styling using SCSS/SASS.

Associate Software Engineer

Mindstorm Studios 07/2021 - 05/2023

Achievements/Tasks

- Focused on system optimization and performance tuning, leading to an extended role assisting senior developers.
- Optimized system performance by addressing bottlenecks in memory management and rendering, improving frame rates by 20% across diverse platforms. Crafted engaging gameplay systems, enhancing user retention and driving a 10% engagement uplift.
- Developed monetization features, including microtransactions and ads, increasing revenue by 15% within the first quarter. Created automated testing pipelines using Jenkins and Selenium, reducing QA cycles by 30% and expediting game release schedules.

Projects

Open Banking Platform Development

Played a key role in creating an open banking solution offering a wide range of financial services to merchants. Integrated with 250+ EEA banks to ensure seamless compatibility and system integration. Built scalable backend infrastructure capable of handling high-volume requests efficiently.

Cryptocurrency Integration for Merchants

Contributed to the development of a platform enabling merchants to accept cryptocurrency transactions, including NFT payments. Enabled users to convert FIAT currency to crypto for purchases or deposits. Designed and developed an intuitive interface for managing crypto transactions, enhancing user experience in tracking, reporting, and ensuring KYC compliance.

ScribeMedix AI Medical Scribing for Athena Health

ScribeMedix is an AI-powered medical scribing tool integrated with athenahealth that automatically records doctor-patient appointments via WebRTC, transcribes conversations using OpenAI Whisper at ~95% accuracy, and auto-fills EHR forms through intelligent API automation reducing physician interaction with the system by up to 70% while generating a structured clinical summary for every patient record.

SaaS-Based CRM Platform

Developed a modern CRM frontend with modules for lead management, task delegation, and analytics. Built a secure multi-tenant architecture to handle organizational data across different clients. Integrated functionalities such as lead tracking, sales pipeline management, task assignment, and performance dashboards. Deployed the platform using Docker containers with PM2 for process management, ensuring high availability and scalability.

AML Watcher Module

Built an AML (Anti-Money Laundering) watcher to monitor transactions for suspicious activity and ensure compliance with regulatory standards. Integrated rule-based alerts and risk scoring to identify high-risk users or patterns in real time, helping prevent fraud and financial crimes.

EDUCATION

BS Computer Science

Information Technology University (ITU) - (2019-2023)

CERTIFICATES

Certification of Completion - MEVN Stack

Training Program, Lahore, Pakistan

Certification of Appreciation - Google Developer Student Club (IEEE Chapter)

Lahore, Pakistan

Certification of Achievement - Mindstorm Studios

Lahore, Pakistan

Award for Best External Advisor

Forman Christian Collage University, Lahore, Pakistan